The Bounded Observer

Operational closure, hyperbolic records and conditional physical applications

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Abstract

We formulate the Universal Hyperbolic Law as an operational framework for bounded normalized readings of positive records. Homogeneity, strong self-duality and a connected equivariant reading–generator correspondence select a simple complex Hermitian matrix algebra. Its elementary binary record has a Lorentz cone, a hyperbolic projective interior and exact rapidity composition under canonical filters. With explicit complex composition and complete positivity premises, records, effects, instruments, purification and environment dilation form a closed operational loop. We prove this closure without demanding that the framework uniquely select every Hamiltonian or constant. A full retained-algebra forgetting law supplies a realizable entropy selector; cumulative dissipative exposure gives a clock-invariant obstruction to finite-resource boundary attainment. We organize exact applications to growth, light, composite records, fields, matter and gravity, identifying the additional physical premises each uses. The singularity hypothesis acquires model-specific regularity and concentration statements. A factorization criterion distinguishes model-dependent preparation readings from claims fixed across physical completions. Historical empirical outcomes and high-scale hypotheses are preserved at their supported strength. The resulting library is mathematically closed as an operational framework with explicit applications, rather than an established derivation of all actual physics from boundedness alone.

I. The question and the object

An observer gathers an amount of evidence and reads a normalized description of it. The amount and the description obey different combination rules. Independent raw records add; a normalized share is a ratio. A ceiling on that ratio does not cap the total amount. This distinction is the starting point of the Universal Hyperbolic Law, abbreviated UHL.

The mathematical object is an operational record system. A state says what readings are possible; an operation says how one record changes into another; a composite says how parts describe a whole. The law is closed in the sense that these objects and operations fit together, and applying them produces another object of the same kind. That meaning does not require a theory of everything or a unique numerical value for every physical parameter.

The eight companion papers carry the detailed proofs and physical applications. This capstone states the common structure, supplies an explicit operational closure theorem and identifies what each application adds. Imported classification, dilation, regularity and thermodynamic theorems are distinguished from deductions proved here and in the companions.

Paper	Title	Main role
1	The Geometry of Bounded Records	Cone classification, trace readings and canonical hyperbolic filters
2	One Motion on the Line	Fixed-rate growth, mixture curvature and conditional response peaks
3	Light Time and Horizons	Event, signal and clock identifications; probability and energy skies
4	Two Observers	Complex composition, purification, retained-algebra forgetting and cumulative rates
5	Where Singularities Can Live	Specified interior regularity, boundary concentration and a fluid conjecture
6	The Partner Freedom	Local-frame connections, conditional matter representations and charges
7	Gravity as Equilibrium	Stationary hyperbolic reduction and conditional horizon thermodynamics
8	The Ceiling and the Constants	State-readout factorization, dynamical selection and physical completion

The central distinction is between a completed operational structure and its physical realization. The framework can be used by specifying a subject's records, operations, retained readings and physical identifications, then deriving what that model permits or excludes.

II. What is required, and what follows

Let V be a finite-dimensional real vector space with a proper cone C: closed, convex, pointed and generating, with nonempty interior. A strictly positive linear normalizer τ defines the compact state base $\Omega = \{x \in C : \tau(x) = 1\}$. Strict positivity prevents an unbounded normalized sequence: its unit directions would converge to a nonzero positive record with zero normalizer.

The foundational premises are individually stated.

H1: homogeneity. Invertible linear cone automorphisms act transitively on strictly positive records.

H2: strong self-duality. A specified positive inner product identifies positive records and positive readings. Choose a compatible equilibrium unit u and its unit-preserving reversible group.

H3: connected reading–generator correspondence. After the Jordan construction, put $V_0 = u^{\perp}$. There is a linear bijection from V_0 to the derivation algebra, equivariant under the connected unit-preserving group K_0, such that the generator assigned to a reading conserves that reading:

$$\phi(kx) = k\phi(x)k^{-1}, \qquad \phi(x)x = 0.$$

The first two premises invoke the Koecher–Vinberg symmetric-cone theorem and the Euclidean Jordan classification [1]. Paper 1 proves that H3 excludes direct sums and selects $\mathrm{Herm}_n(\mathbb{C})$, including the vacuous scalar case. For a nontrivial system,

$$\phi(x)y = \alpha i[x, y], \qquad \alpha \neq 0.$$

The calibration of α remains explicit. Transposition reverses its sign, so connected equivariance is necessary; ordinary equivariance under the entire disconnected Jordan group would exclude the intended complex model. A separate commutator identity can fix $|\alpha| = 1/2$ in the declared Jordan convention.

For a minimal nontrivial binary record, this selects a complex qubit and three Bloch directions. It does not select the dimension of every composite, nor identify those directions with physical spatial axes before an event or signal bridge is supplied. The finite-dimensional, homogeneous, self-dual and reading–generator premises have real content. Boundedness alone also permits classical simplices and many other models.

Continuous additive positive readouts of raw complex records have the trace form

$$\ell_E(M) = \mathrm{Tr}(EM), \qquad E \succeq 0.$$

Complementary readings whose sum is the total amount give $0 \preceq E \preceq I$ and probability $p_E(\rho) = \mathrm{Tr}(E\rho)$. Effects separate states. A record is therefore sufficient for the declared future operation class precisely when equal records give equal future readings. If an experiment depends on omitted information, the record must be enlarged.

III. The exact hyperbolic law

A binary raw record is

$$M = x_0 I + \mathbf{x} \cdot \sigma, \qquad \det M = x_0^2 - |\mathbf{x}|^2.$$

Positivity is $x_0 \geq |\mathbf{x}|$. Normalization gives the Bloch ball, and its Hilbert projective metric is twice the conventional curvature-minus-one hyperbolic distance. Along a fixed diameter put $b \in (-1, 1)$. Canonical positive linear filters multiply the two odds, so

$$r = \mathrm{artanh}\, b = \tfrac{1}{2} \log \frac{1+b}{1-b}, \qquad b_{12} = \frac{b_1 + b_2}{1 + b_1 b_2}.$$

This is the exact UHL composition in its declared coordinate and operation class. Aczél's general associative representation allows other monotone additive charts; the linear positive-record/filter structure is what chooses this one. An additive amount such as energy, momentum or population is not automatically the bounded b.

The determinant-one congruence action gives $SL(2, \mathbb{C}) \to SO^+(1, 3)$ and faithful projective action $PSL(2, \mathbb{C})$. Non-collinear canonical boosts have polar rotations; for a geodesic hyperbolic triangle their holonomy angle is its oriented area. Such identities are exact within the cone geometry. Their use as velocity addition or spacetime holonomy additionally identifies the cone with physical signals or momenta.

On the line, fixed diagonal filters include exponential growth, two-level thermal weights and exact all-or-none binding models. Under a declared mixture clock s, fixed-rate populations satisfy

$$\frac{d^2}{ds^2} \log \frac{A(s)}{B(s)} = \mathrm{Var}_A(r) - \mathrm{Var}_B(r).$$

The curvature is a difference of hidden variances. Homogeneity is sufficient for a straight log-odds line, but not necessary because the variances can cancel. With $ds/dt = a(t)$, laboratory curvature

also has a clock term $a'\Delta\bar{r}$. This supplies a useful identity and an identifiability limit together. Paper 2 proves the corresponding fixed-intervention and response-peak laws.

IV. How the observer closes as a loop

We now declare complex tensor-product composition and the usual operational requirement of complete positivity: an allowed local branch must remain positive when applied beside an arbitrary finite ancillary record. These are composition premises, not consequences of two isolated cones.

For each finite complex Hilbert space H, use density matrices as states and $0 \preceq E \preceq I$ as effects. An instrument has finitely many completely positive, trace-nonincreasing branches $\mathcal{J}_j$ whose sum is trace preserving. If outcome j occurs, its probability and normalized output are

$$p_j = \operatorname{Tr} \mathcal{J}_j(\rho), \qquad \rho_j = \mathcal{J}_j(\rho)/p_j \quad (p_j > 0).$$

Theorem 1: operational closure. The class of all finite complex record systems with these states, effects and instruments is closed under serial operations, parallel composition, probabilistic mixing, selective readings and discarding parts. Every state can be obtained as the marginal of a pure joint state, and every trace-preserving operation can be implemented as a reversible operation on a larger system followed by discarding an environment.

Proof. Positivity and trace normalization are preserved by channels and partial traces. Compositions and tensor products of completely positive maps are completely positive; trace conditions compose. Sequential instrument branches $\mathcal{J}_k\mathcal{J}_j$ form another instrument because their total is the composition of the two channels. Convex mixing preserves the same conditions, and normalization of any positive branch with nonzero trace gives a state.

For a spectral decomposition $\rho = \sum_i \lambda_i |i\rangle\langle i|$, the vector $\sum_i \sqrt{\lambda_i}|i\rangle|i\rangle$ purifies it. A finite channel has a Kraus representation $\Phi(\rho) = \sum_j K_j \rho K_j^\dagger$ with $\sum_j K_j^\dagger K_j = I$. The map $V\psi = \sum_j K_j\psi \otimes |j\rangle$ is an isometry, and tracing its environment gives Φ. After padding finite spaces, the isometry extends to a unitary on a system plus a prepared environment. This is the finite-dimensional dilation theorem [2]. The outcome register similarly realizes instruments. Every construction remains in the same finite complex class. □

The loop is explicit: a whole has a state; a part reads its marginal; a reversible whole operation becomes a possibly irreversible part operation; its next state and readings remain admissible. Pure wholes can make mixed parts. This is a mathematically complete meaning of closure. The individual ingredients are established quantum information; the theorem identifies exactly which ingredients and premises complete the bounded-observer loop.

For two binary parts, a pure amplitude matrix Ψ gives $P = \Psi\Psi^\dagger = \frac{1}{2}(p_0 I + \mathbf{p}\cdot\sigma)$ and

$$p_0^2 - |\mathbf{p}|^2 = 4|\det\Psi|^2, \qquad C = \frac{2|\det\Psi|}{p_0}, \qquad C^2 + |\mathbf{p}/p_0|^2 = 1.$$

If P is also calibrated as particle four-momentum, the cone mass is $2|\det\Psi|$ and $C = 1/\gamma$. This is an exact correspondence, not a proof that all actual mass is entanglement. A photon's null momentum and possibly entangled polarization occupy different record spaces. A mixed environment can also implement forgetting through classical correlations; a pure-environment dilation is always possible but is a description choice.

V. Selection and time without a hidden clock assumption

Let $\mathcal{A} \subset M_d(\mathbb{C})$ be the complete unital algebra of retained readings, closed under adjoints, and $E_{\mathcal{A}}$ its trace-preserving conditional expectation. The channel family

$$\Phi_u = E_{\mathcal{A}} + e^{-u}(\mathrm{id} - E_{\mathcal{A}}), \qquad u \geq 0,$$

forms a semigroup and selects $\rho_* = E_{\mathcal{A}}\rho_0$. For every compatible σ,

$$D(\sigma\|\rho_*) = S(\rho_*) - S(\sigma) \geq 0.$$

Thus forgetting to a complete retained algebra is a realizable unique maximum-entropy selector. The retained algebra and readings are specified data. A Gibbs selector retaining only a few means is generally nonlinear in the initial state and cannot be the universal limit of one fixed linear channel. Paper 8 gives a three-level counterexample and separates this dynamical issue from the entropy optimization theorem [3,4].

An external clock is unnecessary for an additive exposure parameter u, but physical elapsed time needs a clock rule. For a well-defined finite-dimensional time-local GKSL model [4], let $G(t) = \sum_k L_k^\dagger L_k$. A strictly positive initial state obeys

$$\lambda_{\min}(\rho(T)) \geq \lambda_{\min}(\rho(0))e^{-\Xi(T)}, \qquad \Xi(T) = \int_0^T \|G(t)\|_{\mathrm{op}} dt.$$

The no-jump propagator supplies a positive contribution whose smallest singular value is at least $e^{-\Xi/2}$. This proves the bound. Under a differentiable common clock change with strictly positive derivative, the jump amplitudes transform by the square root of the derivative, making the integrated exposure invariant for the specified representation.

Finite exposure prevents interior-to-boundary arrival. Finite clock duration alone does not. A finite-time swap with a pure environment can purify a part while its effective local time-local exposure diverges. Infinite rapidity also can occur in finite assigned time under an unbounded driving rule. Every claim about time must therefore name its clock and integrated resource, rather than silently replace steps by seconds.

Even after eliminating a common clock, dynamics need not be selected by equilibrium. Two locally unitary-covariant reset processes on $\mathbb{C}^2 \otimes \mathbb{C}^3$ can share the same unique state $I_6/6$ and conserved algebra while their relative rates differ. Plotting one logarithmic polarization loss against the other reads that ratio without an external clock. Full $PU(d)$ covariance of the actual dynamics on one carrier is a stronger condition: it selects $\mathcal{L} = \kappa(D - \mathrm{id})$, leaving one common calibrated rate. This is an exact selector when that premise applies.

VI. Light, events and horizons

To turn record geometry into spacetime geometry, specify a local invertible linear event map $E_x : T_x\mathcal{M} \to \mathrm{Herm}_2(\mathbb{C})$ taking future displacements to positive records and null signals to rank-one records. Then $g_x(v, v) = \det E_x(v)$ is Lorentzian. The cone fixes its conformal class. A tangent-only, degree-one, Lorentz-invariant calibrated clock gives proper time proportional to $\sqrt{g(dx, dx)}$.

This derives the metric and clock form after the event and clock bridges. It does not derive global flatness or an equivalence principle from an abstract qubit. For signal wave covectors and normalized observer velocities, the Lorentz pairing gives Doppler shift and the Busemann reading $B_\xi(U) = \log\langle U, (1, \xi)\rangle_L$.

Two distinct sky measures illustrate why operational meanings matter. With $\mu = \cos\vartheta$ and boost β, the equivariant harmonic probability is

$$P_2(\mu) = \frac{1 - \beta^2}{4\pi(1 - \beta\mu)^2},$$

whereas normalized bolometric energy for isotropic-rest radiation is

$$P_4(\mu) = \frac{3(1-\beta^2)^3}{4\pi(3+\beta^2)(1-\beta\mu)^4}.$$

The second has the radiation M1 energy moments; the first does not. Probability averaging, conformal probability barycentres and energy rest frames cannot be interchanged.

Within flat massless scalar wave models, sharp light-cone retarded support together with positivity as a measure selects three spatial dimensions. Higher odd dimensions can also have sharp support, but their retarded distributions are not positive measures. That restricted result does not select an arbitrary physical field operator.

An arbitrary observer path need not be a geodesic. A specified isotropic independent random-filter model has asymptotic geodesic tracking and drift $\mathbb{E}[\rho \coth \rho - 1]$ under its integrability hypotheses. In a physical one-direction accelerated trajectory, an acceleration horizon instead requires convergence of $\int^{\infty} e^{-\eta(\tau)} d\tau$. Velocity-space geometry alone does not decide that integral. Paper 3 gives the signal, clock and thermal-amplitude scopes in full.

VII. Fields and the singularity hypothesis

A bounded field value can still have an unbounded gradient. A field result must therefore state the source, target metric, dynamics, initial data and energy convention. The Singularity Hypothesis, SIH, has a precise conditional form in the following models.

For hyperbolic harmonic-map heat flow from a fixed closed flat source, smooth initial data have global smooth evolution and non-increasing maximum radius and energy density (Paper 5, Theorem 3: radius and Bochner bounds, with local heat-flow theory [5]). The radius bound supplies the compact target region needed for continuation; the target itself need not be compact. Isotropic local first-derivative quadratic dissipation invariant under all target isometries has the hyperbolic Dirichlet form, provided its constitutive coefficient is uniform and no extra fields or derivative terms are allowed.

For the conventional mixed-qubit Bures hemisphere, a positive initial mixedness margin gives global regular heat flow on the same fixed flat source through a weighted maximum-principle estimate. The Bures and hyperbolic metrics are different: purity lies at finite distance in one and infinite distance in the other.

In two-dimensional round-sphere heat flow, concentration can produce harmonic-sphere bubbles. With energy $\frac{1}{2}\int |dn|^2$ and unit target sphere, a degree-one bubble costs 4π [6]. The pure-qubit Fubini–Study radius gives π instead. Bubbling, local rescaling and topology changes of a limiting weak description are not statements that every edge field must blow up or that every singularity is exactly one Lorentz boost.

Specified mixing and intensity-zero constructions can regularize constrained pure descriptions. Imported fixed-background hyperbolic wave-map and equivariant Schrödinger-map theorems have separate dimension, symmetry and function-space assumptions. These conclusions cannot be extended to arbitrary dynamics merely because its state values lie in a bounded interior.

The revised hyperbolic-friction fluid has an explicit optional incompressibility and elliptic-pressure closure, with global regularity conjectural. Rejecting a flat constitutive ruler in that chosen model does not solve its replacement PDE or an unrelated Navier–Stokes problem. Schwarzschild curvature divergence in finite freely falling proper time also lies outside these fixed-source target-map theorems. Paper 5 preserves both the proven SIH pattern and its limits.

VIII. Matter and gravity as physical applications

An internal frame that is a genuinely local redundancy of all physical operations admits a connection. Making that connection dynamical requires an action. A purification partner's freedom preserves one marginal but can affect joint readings, so it is not automatically a universal physical gauge redundancy.

A small partner loop gives a centred generator covariance. At maximal mixing,

$$K_{ab} = \frac{1}{n}\,\mathrm{Tr}(T_a T_b) - \frac{1}{n^2}\,\mathrm{Tr}(T_a)\,\mathrm{Tr}(T_b).$$

The identity generator has zero response. Abelian charge comparisons need relative charges and a reference. Lorentz invariance, gauge invariance, locality and a declared leading parity-even curvature-quadratic action yield a Yang–Mills form, while distinct simple factors can keep independent coefficients.

For a specified chiral $SU(N)$ family, one Higgs doublet, stated Yukawa terms and no right-handed neutrino, anomaly cancellation gives $Y_Q = Y_H/N$ and $Y_L = -Y_H$. At $N = 3$ the familiar charges follow. The global $SU(2)$ condition is $n_g(N+1)$ even [7], giving odd N when n_g is odd. Neither three colours nor three generations follows from this arithmetic alone. A zero bare-mass premise also permits vectorlike Yukawa models and therefore does not by itself force chirality. A right-handed neutrino permits a $B - L$ charge freedom without requiring it to be gauged.

Paper 6 gives exact stabilizer and charge statements with their representations declared. It also constructs a zero-electric-charge conjugate-pair ensemble whose local electric-to-colour covariance ratio is two. Changing the boundary constraints gives different ratios. Equating such a covariance with an actual kinetic coupling needs a response-to-action mechanism; it is not automatic coupling unification.

Gravity has two distinct routes. Stationary vacuum Einstein equations on a timelike Killing patch reduce to an Ernst map into a hyperbolic plane coupled to a three-dimensional quotient geometry [8]. This route begins with Einstein's equation. For Kerr, the chosen stationary Killing field becomes null at the ergosurface, generally distinct from the event horizon. Smooth patch reconstruction does not remove genuine curvature singularities.

The thermodynamic route instead starts with area entropy $S = \eta A$ in nats, Unruh temperature, local horizon equilibrium, Clausius balance for every local horizon and conserved stress tensor. Jacobson's argument yields Einstein's equation and

$$G = \frac{c^3}{4\hbar\eta},$$

leaving Λ free [9]. The area law is a physical premise; locality and bounded cells can also allow volume-law states. An independently imposed non-collapse condition gives a conditional effective density ceiling [10]. A de Sitter horizon repackages a supplied Λ as its entropy capacity; it does not predict that value or settle vacuum-energy renormalization.

An equation of state does not determine microscopic quantum fluctuations or non-equilibrium evolution. Such an extension needs explicit variables, constitutive laws, constraints and initial data. The equilibrium derivation neither proves that quantization is unnecessary nor supplies a well-posed non-equilibrium theory by itself. Paper 7 states these distinctions alongside the exact stationary and cosmological identities.

IX. What is fixed, and what has been tested

For fixed constraint and readout operators, a unique maximum-entropy state gives

$$\rho_*(q) = Z^{-1} e^{-\sum_i \beta_i(q) Q_i}, \qquad c_a(q) = \mathrm{Tr}(R_a \rho_*(q)).$$

On the full-rank relative-interior domain, these responses are smooth and their Jacobian rank is at most the number of independent constraints modulo identity. More outputs than constraints yield relations, not necessarily individually fixed outputs. A state-independent output is one constant throughout the declared domain. The full retained-algebra selector in Part V realizes an especially concrete version of this theorem.

Theorem 2: physical completion criterion. Let $\mathfrak{M}$ be a nonempty class of admissible physical completions after operational gauge identification. For an independently specified data map B and observable X, the law fixes X exactly when it is constant on $\mathfrak{M}$. It determines X from those data exactly when $X = F \circ B$, equivalently when X is constant on each fibre of B.

Proof. Factorization gives equal outputs for equal data. Conversely, define $F(b)$ by any model in the fibre; fibre constancy makes the definition independent of the choice. □

This theorem makes both closure and further derivation testable. Calling a number boundary data is legitimate after supplying its map and factorization. It does not locate the data on a physical horizon or choose a generator, representation or physical readout. A fixed finite deterministic processor also cannot exactly program a continuum of inequivalent unitary gates using nonorthogonal finite program states [11]. Paper 8 proves that conditional restriction and explains the mechanisms that escape it.

The high-scale Higgs conditions $\lambda = \beta_\lambda = 0$ remain a separate criticality hypothesis. Neither maximum entropy nor stationarity in a bounded scale coordinate proves them. The historical 171.2 ± 0.4 GeV top-pole target is retained as a dated proposal, not a certified simultaneous fixed-cell precision prediction. An algebraic inversion of the existing Buttazzo et al. approximate stability fit at declared inputs gives about 170.997 GeV; that is a conditional use of the published fit, not a new parameter-free prediction [12]. A consistent spectrum, matching scheme, loop order, mass definition and input covariance are required before a precision test.

Historical numerical tests also retain their original status. The accompanying provenance records distinguish completed analyses from scripts newly rerun for this revision.

Item	Supported status
T15 original intervention test	Reported failed registered criterion; later clock correction exploratory
T15b restricted intervention test	Reported missed registered bars; no retrospective reclassification
T20 registered BAO plus supernova comparison	Neutral primary result, reported $\Delta\chi^2 = -3.37$
T20 later CMB extension	Exploratory extension disfavoured saturation, reported $\Delta\chi^2 \simeq +117.46$
Common-coupling rule	Additional hypothesis with reported weak-angle discrepancy
Residual electric/strong alignment	Post-hoc observation; precision crossing and chance significance not certified here
Hormesis universal necessity and $2D_a$ peak	Withdrawn at that generality; exact conditional replacements in the amendment

The empirical analyses in this table have not all been independently rerun in the present revision. Exact proof checks are supplied separately and are not empirical validation. Recognizable structures across several applications also do not define a probability that the entire programme occurred by chance. No unique UHL prediction is claimed here to have passed a previously blinded physical test.

X. What UHL is, and how to use it

UHL, as finalized here, is an operational geometry of bounded normalized records, their positive filters and their lawful composition. Its root classification has explicit strong premises. Its binary projective geometry is hyperbolic; its complex operations and composites form the closed loop of Theorem 1. Its state-selection and boundary-attainment statements specify retained information and cumulative resources. SIH is the regularity/concentration pattern proved for named field models within that organization.

That is a complete framework in the operational sense requested: it has defined objects, defined operations, a closed composition rule, exact deductions and a criterion for its physical completions. It has further work because applications can choose dynamics, preparations and physical bridges. Closure does not make every such choice unique.

To apply it to a new subject, identify the positive raw record and normalizer; specify the readings and intervention class; state the composition and retained algebra; declare the clock or clock-invariant resource; provide the physical readout and any action or field equation. Then derive the permitted responses. The same-fibre criterion shows whether a proposed parameter follows, is data-dependent, or still needs an additional model premise. A countermodel can decide that last issue exactly.

The scientific contribution is an explicit common organization with proved identities and usable conditional laws. Many ingredients are established mathematics and physics; their appearance does not make them independent empirical discoveries. The library's significance can be judged by the classification proof, the scope of its exact applications, and future discriminating tests. Mathematical coherence is established at those scopes; universal physical validity remains a scientific question.

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